Differential CC $1\mu1\pi^{\pm}$ cross sections on argon in the NuMI beam

Inclusive measurement and the proton-tagged hadronic-mass / TKI subsample

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$\nu_\mu + \bar{\nu}_\mu$ charged-current single charged pion • FHC, RHC, combined

Outline

Part I, Inclusive CC$1\mu1\pi^{\pm}Xp$

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2. Beam, exposure, selection
3. Cut-flow & performance
4. Unfolding & response
5. Differential results + generators
6. Integrated & total cross sections
7. Systematics & sideband validation

Part II, Proton-tagged (W / TKI)

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3. Observables & closure
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5. Background composition

Part III, Multi-pion (2π, 3π)

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Backup

Part I, Inclusive CC $1\mu1\pi^{\pm}Xp$

Signal definition

$\nu_\mu/\bar{\nu}_\mu$ charged-current, exactly one muon and one charged pion, any number of protons, no π^0/kaons/heavier mesons; muon $p_\mu > 0.15$, pion $p_\pi > 0.175$ GeV/c, in the fiducial volume.

- Signal defined by the final state $\Rightarrow$ every interaction mode contributes, via primary pion production *and* via FSI:

Mode	% of signal	Pion origin
RES ($\Delta(1232)$, higher N^*)	77.5	primary
DIS / SIS	17.5	primary (higher W)
QE	2.6	FSI only (no primary π)
COH	2.1	primary, low $\lvert t\rvert$
MEC / 2p2h	0.3	FSI only

- FSI also migrates events both ways: absorption/charge exchange turns multi-π into 1π, and 1π into 0π.
- NuMI off-axis: ν_μ-dominant FHC, $\bar{\nu}_\mu$-enriched RHC. Observables: p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$.

Exposure (fake data, blind)

FHC (Runs 1,2,4,5)	8.86×10^{20} POT
RHC (Runs 1,2,3,4)	1.11×10^{21} POT
Combined	1.99×10^{21} POT

Event selection: cut sequence

A CC$1\mu1\pi^{\pm}Xp$ topological selection: one muon, exactly one contained charged pion, any number of protons. Applied in order (cosmic rejection first, then PID, then kinematic cleanup):

1. **NeutrinoSlice**, ≥ 1 track, software trigger.
2. **InFiducialVol**, SCE-corrected ν vertex in FV.
3. **Topological**, Pandora score > 0.67; primary cosmic rejection (EXT $\times 33$ down).
4. **MuonCandidate**, ≥ 1 primary track passes μ PID; longest $= \mu$.
5. **ContainedPion**, exactly one other track passes π PID, endpoint contained; highest LLR $= \pi$.

6–7. μ/π **In3Planes**, hit on each of U, V, Y.

8. **ShowerCut**, veto primary EM showers (π^0 rejection; one soft shower tolerated).
9. **OpeningAngle**, $\theta_{\mu\pi} < 2.6$ rad (149°); rejects large-angle cosmic/dirt.
10. **Nonprotons** (final), < 4 non-proton-like tracks (LLR> 0.1) and < 5 total primary tracks.

Per-run POT scaling; EXT (beam-off) supplies data-driven cosmics; dirt from the GENIE NuMI overlay. **Final: efficiency** 15.6%**, purity** 55% (measured phase space).

Particle identification (muon & pion candidates)

Track-level PID combines Pandora track score, LLR calorimetric PID, Bragg-peak likelihoods and dedicated MIP/pion BDTs:

Muon candidate (Cut 4)

Variable	Threshold
Track score	> 0.80
Vertex distance	≤ 4.0 cm
Track length	> 10.0 cm
LLR PID	> 0.20
MIP BDT	≥ -0.10

Pion candidate (Cut 5)

Variable	Threshold
LLR PID	> 0.10
Vertex distance	< 4.0 cm
Track length	> 20.0 cm
MIP BDT	> -0.10
Pion BDT (vs. p)	> -0.10
Bragg-pion score	≥ 0.08
Endpoint contained	yes

Momentum reconstruction on the next slide.

Momentum reconstruction: range, MCS and the proton range-KE

All three tracks are range-based where possible: a contained track's length maps onto momentum through the particle's dE/dx, which is far more precise than calorimetry at these energies.

Particle	Estimator	PeLEE branch	Note
μ (contained)	range momentum	`trk_range_muon_mom_v`	37% of selected
μ (exiting)	MCS momentum	`trk_mcs_muon_mom_v`	63%, range invalid
π	range momentum, μ hypothesis	`trk_range_muon_mom_v`	m_μ close to m_π
p	range KE, p hypothesis	`trk_energy_proton_v`	$\to$ momentum, see below

p_μ is therefore a hybrid quantity (`..._mom_reco` switches on muon containment) and is MCS-dominated: NuMI muons are energetic and the TPC is small, so most leave it.

Proton: range KE $\to$ momentum

The proton branch gives a *kinetic energy*, so it is converted:

$$|\vec{p}_p| = \sqrt{T_p^2 + 2\, T_p\, m_p}$$

Threshold $|\vec{p}_p| > 0.3$ GeV/c ($T_p \gtrsim 48$ MeV), below this the proton is too short to track. Performance: -9% bias, 21% resolution.

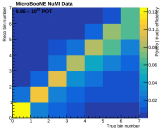
Why the pion uses the μ hypothesis

$m_\mu = 105.7$ MeV, $m_\pi = 139.6$ MeV: the masses are close, so the range$\to$momentum curve is a good proxy for a stopping pion.

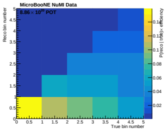
It also returns a momentum, directly comparable to the true p_π it is binned against; which keeps the migration matrix near-diagonal.

Valid only for contained tracks; the selection requires pion containment.

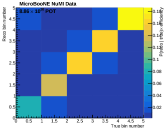
Response (smearceptance) matrices, FHC



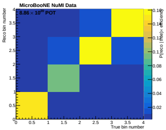
p_μ



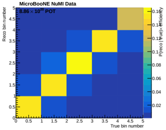
p_π



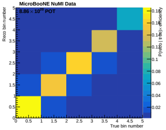
$\cos\theta_\mu$



$\cos\theta_\pi$



$\theta_{\mu\pi}$

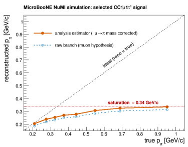


θ_μ

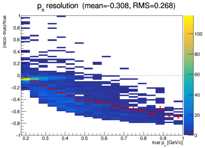
$P(\mathrm{reco}\,|\,\mathrm{true})\times$ efficiency; columns are true bins, rows reco bins. p_μ and the three angular observables are near-diagonal, note p_μ is 63% MCS-reconstructed (exiting muons), whose poorer resolution sets its width. p_π is the outlier, reco piles into the first bin from every true bin, since pions interact hadronically before stopping so a range-based momentum under-estimates p_π. With 50–99% bin migration overall a plain matrix inversion is unusable, hence Wiener-SVD regularisation; at the cost of the A_C additional smearing.

Planned scope change: measured migration diagonals for p_π are 99/24/13/6/6%; the BNB CC1π criterion for a usable bin is $> 68\%$, so only the first passes. A golden-pion BDT was trained and reproduces the BNB classifier performance (43$\rightarrow$65% golden at 45% efficiency) but cannot fix it: even a perfect truth-level golden selection leaves bins 2–5 at 36/24/14/9%. We therefore expect to withdraw p_π from the inclusive selection and quote it on the golden-pion-enhanced and/or proton-tagged samples instead. All other observables are unaffected.

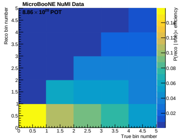
The pion momentum cannot be measured well, and why



What we measure stops responding to the true momentum above ~ 0.35 GeV/c



Per event: right at low momentum, then increasingly too low



Whatever the true momentum (columns), events land in the lowest measured bin

% landing in the correct momentum bin

old 5-bin scheme, true momentum low $\rightarrow$ high					
all pions	99	24	13	6	6
best classifier	99	33	25	15	9
even cheating with truth	99	36	24	14	9
adopted 2-bin scheme: $[0.175, 0.205]$ + everything above					
all pions	97	62			

A bin is normally called usable above 68%. Merging cannot create resolution, it stops us quoting five numbers when we have about one and a half. Three bins was tested and is worse (97/38/39).

Closure $\chi^2 = 0.15/2$ FHC (was $1.96/5$); all three beams overleaf.

- Momentum comes from how far the particle travels, fine for muons, which just slow down and stop.
- Pions don't: most hit a nucleus and vanish part-way, so the track ends early and we read too little momentum. Faster pion, sooner it happens.
- We tried keeping only pions that did stop cleanly, spotted from the track shape by a machine-learning classifier, as good as the published BNB one.
- Still not enough: at equal fraction kept, cheating with truth does no better; the classifier is already at the ceiling.

$\Rightarrow$ pion momentum will be reported only where it is meaningful, everything else we measure is unaffected.

The rebinned pion-momentum result: all three beams

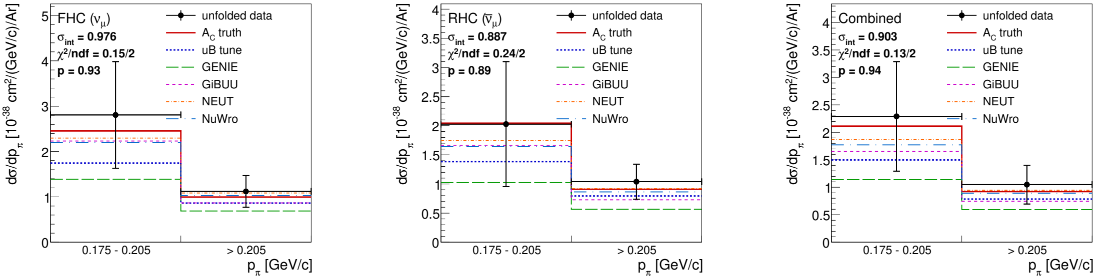
$d\sigma/dp_\pi$ [10^{-38}cm^2/(GeV/c)/Ar]

	FHC		RHC		Comb	
	b1	b2	b1	b2	b1	b2
data	2.81	1.12	2.03	1.04	2.29	1.05
truth	2.46	1.00	2.04	0.91	2.11	0.92
uB tune	1.75	0.86	1.38	0.80	1.50	0.79
GENIE	1.39	0.69	1.02	0.57	1.14	0.59
GiBUU	2.23	0.87	1.67	0.73	1.66	0.75
NEUT	2.30	1.09	1.74	0.92	1.87	0.95
NuWro	2.21	1.03	1.64	0.86	1.77	0.89
σ_{int}	0.976		0.887		0.903	
$\chi^2/2$	0.15		0.24		0.13	

b1 = [0.175, 0.205], b2 = everything above. All curves A_C-smeared.

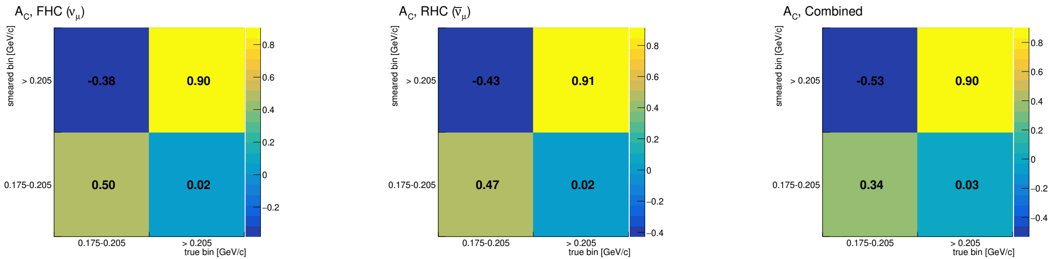
- All three close well: $\chi^2 = 0.15$, 0.24, 0.13 on two bins ($p \geq 0.89$), against 1.96/5 for the old FHC five-bin scheme.
- Unfolded/truth is 1.13 in all three beams, the usual Wiener-SVD offset, now identical across beams where the five-bin scheme was not.
- Generators bracket the data everywhere, NEUT closest, GENIE lowest, so the ordering is stable against beam and binning.
- Predictions use the corrected FTE files; the old ones truncated the p_π tail and were $\sim$8% low.

The same result, drawn



- Data (points, full systematics) against A_C-smeared truth, tune and the four generators, so everything is compared in the same space.
- Bins drawn equal-width for legibility – they are 0.030 and 0.795 GeV/c wide, so to scale the first would be 4% of the axis. Area is therefore not proportional to cross section; σ_{int} is printed on each panel instead.
- The uncertainty on bin 1 is large (42–53% depending on beam, against 29–34% on bin 2): it is a 30 MeV/c slice, and that is the price of a bin narrow enough to be resolved.

Why the truth curves are smeared before comparison



- Wiener–SVD returns $\hat{x} = A_C t$, not t: the measurement is a smeared version of the truth, so any prediction must be passed through A_C before it can be compared with it.
- A_C is not normalisation-preserving – rows do not sum to one – and the negative entry (true bin 1 into smeared bin 2) is the regularisation borrowing between the two bins: they are not independent even after merging.
- FHC and RHC regularise similarly (diagonal 0.50, 0.47); the combined fit borrows appreciably more (diagonal 0.34, off-diagonal -0.53). Despite that, unfolded/truth is 1.13 in all three – the offset is a normalisation effect and does not track how hard A_C smears.

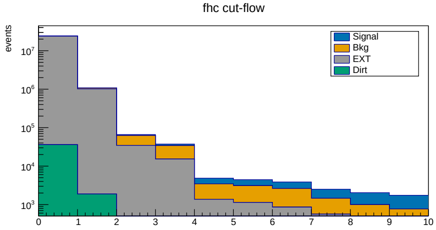
Inclusive cut-flow, FHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	5 580	279 180	23.92M	100.0	0.02
InFiducialVol	4 632	62 146	1.02M	83.0	0.42
Topological	3 501	28 040	34 481	62.7	5.2
MuonCandidate	3 143	18 980	15 206	56.3	8.3
ContainedPion	1 395	2 094	1 348	25.0	28.5
MuonIn3Planes	1 336	1 975	1 122	23.9	29.9
PionIn3Planes	1 239	1 770	854	22.2	31.9
ShowerCut	1 048	898	558	18.8	41.6
OpeningAngle	1 040	813	186	18.6	50.9
Nonprotons	**974**	**643**	**128**	**17.5**	**55.7**

MC prediction, per-run POT scaled; data blind. Dirt $\lesssim 1$ event at final cut.

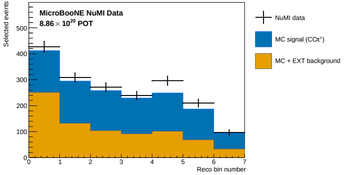
Final cut (Nonprotons), all beams:

Beam	Signal	Bkg	EXT	Eff%	Pur%
FHC	974	643	128	17.5	55.8
RHC	1 081	746	133	17.6	55.1
Comb	2 055	1 389	261	17.5	55.4

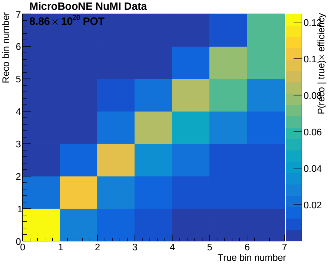


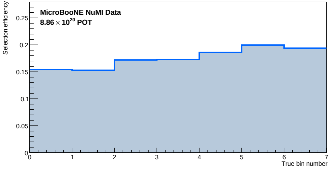
FHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale). Two big drops: Topological (cosmics) and ContainedPion (single-π topology).

Reconstruction, response & efficiency



Reco p_μ (data/MC stack)



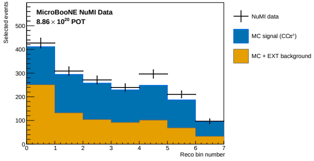


Efficiency vs p_μ

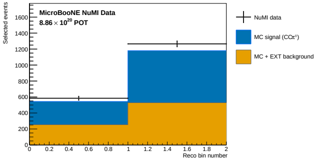
Response matrix

Wiener-SVD unfolding (framework `WienerSVDUnfolder`, 2nd-derivative regularisation) with the full systematic covariance from the universe throws.

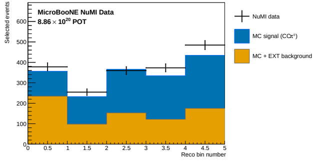
FHC: selected events (POT-scaled), signal vs background



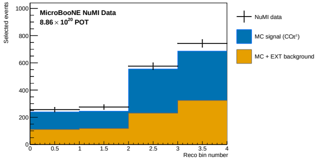
p_μ



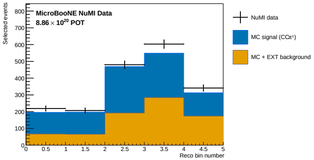
p_π



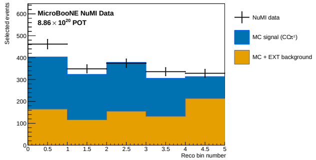
$\cos\theta_\mu$



$\cos\theta_\pi$



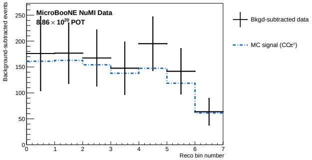
$\theta_{\mu\pi}$



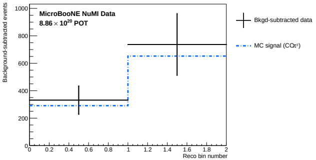
θ_μ

Flat reco-bin space, scaled to full FHC exposure. Points: (blind) fake data; stack: MC signal + background. Purity $\simeq 55\%$ at the final cut.

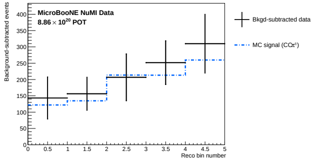
FHC: after background subtraction



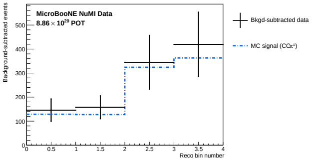
p_μ



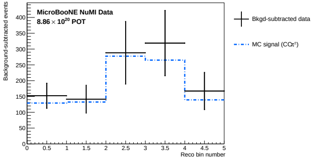
p_π



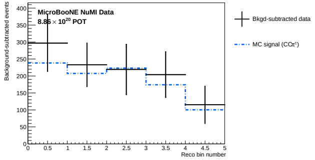
$\cos\theta_\mu$



$\cos\theta_\pi$



$\theta_{\mu\pi}$

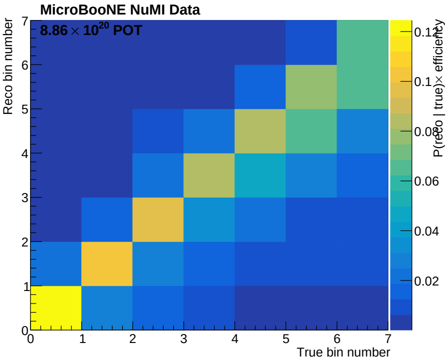


θ_μ

Data − estimated background vs CV MC signal; the input to the unfolding. Background from MC (ν) + beam-off EXT (cosmic) + dirt overlay; validated by the sidebands.

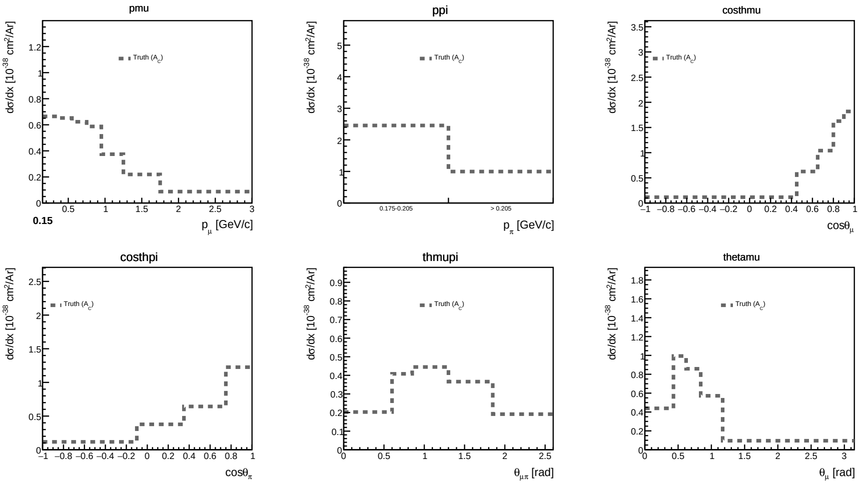
FHC: unfolding

- The background-subtracted reco spectrum is not the cross section: it still carries the detector response and the selection efficiency.
- Unfolding inverts the response matrix A_{ij} shown earlier. With 50–99% bin migration a plain inversion is unusable, so the analysis uses Wiener-SVD regularisation.
- The price is the additional-smearing matrix A_C: the result is $A_C \times$(true signal), so every model must be multiplied by A_C before it is compared to the data.
- The next slides build up that comparison one curve at a time.



response matrix, p_μ

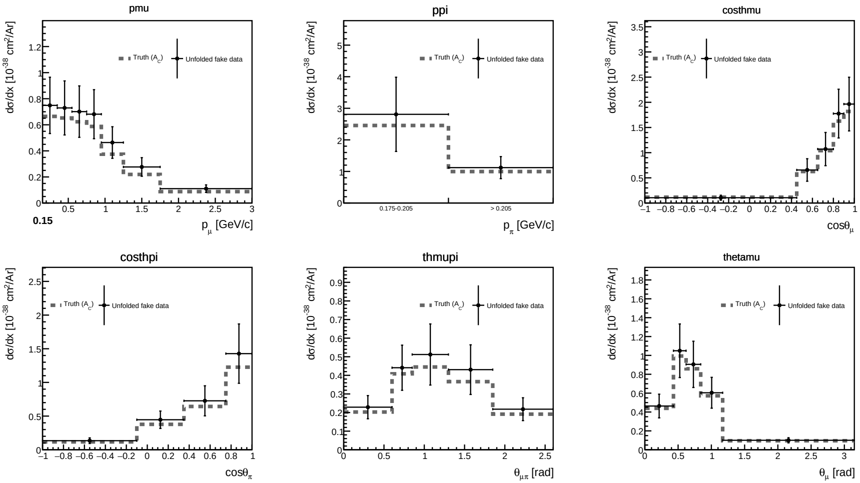
recover



Poisson-thrown fake data are generated from the MicroBooNE tune; after unfolding we should recover this A_C-smeared truth.

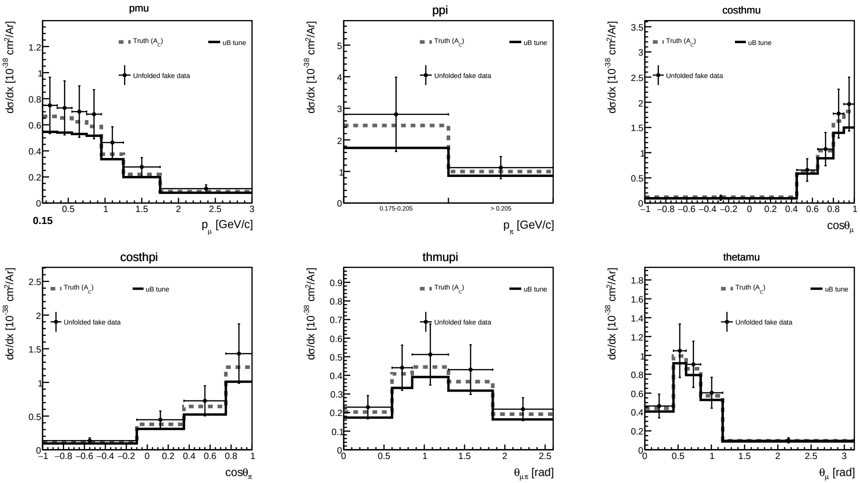
p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV$/c$).

Differential cross sections, FHC: + unfolded fake data



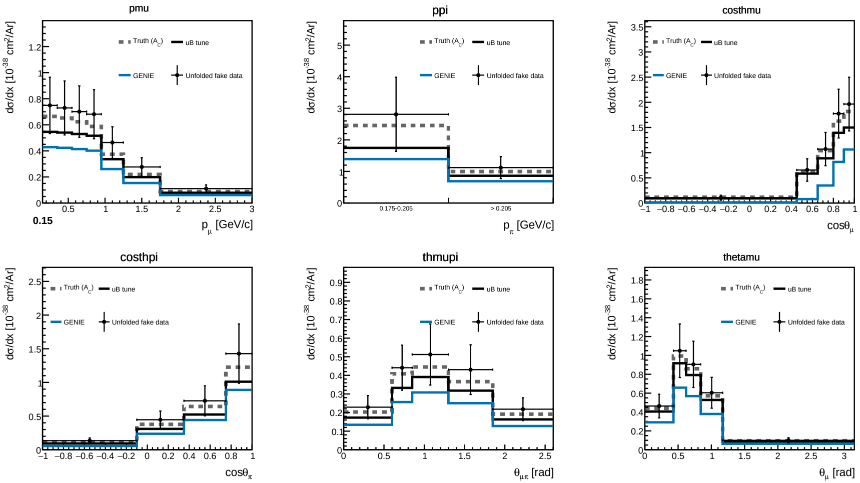
The unfolded points sit on the truth: the response, background subtraction and normalisation close. p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV/c).

Differential cross sections, FHC: + MicroBooNE tune



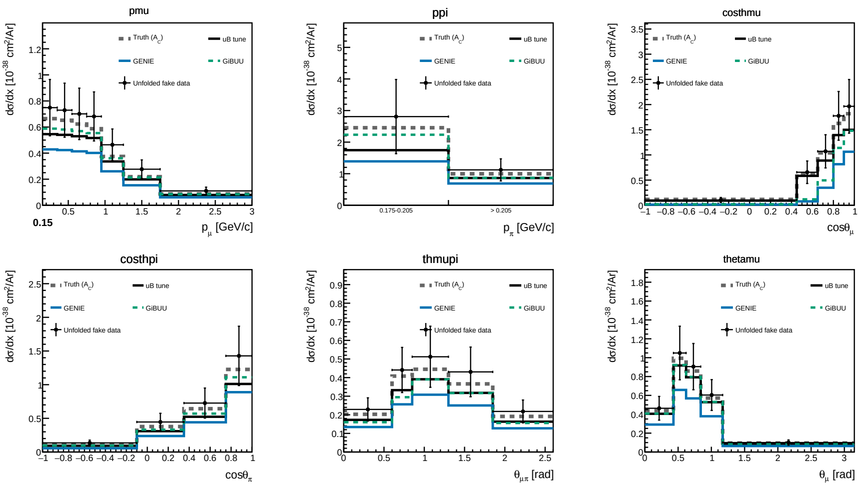
The tune is the central-value model the fake data were thrown from. p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV$/c$).

Differential cross sections, FHC: + GENIE



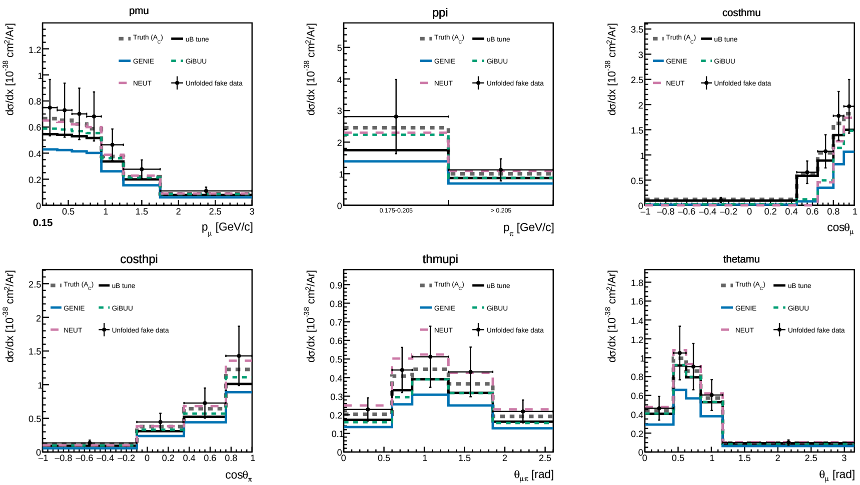
External generators enter, each A_C-smeared so it lives in the same space as the measurement. p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV/c).

Differential cross sections, FHC: + GiBUU



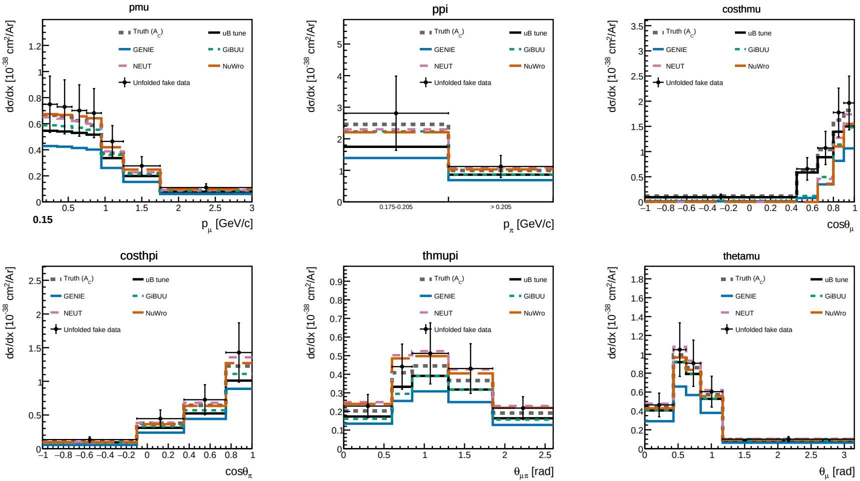
p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV$/c$).

Differential cross sections, FHC: + NEUT



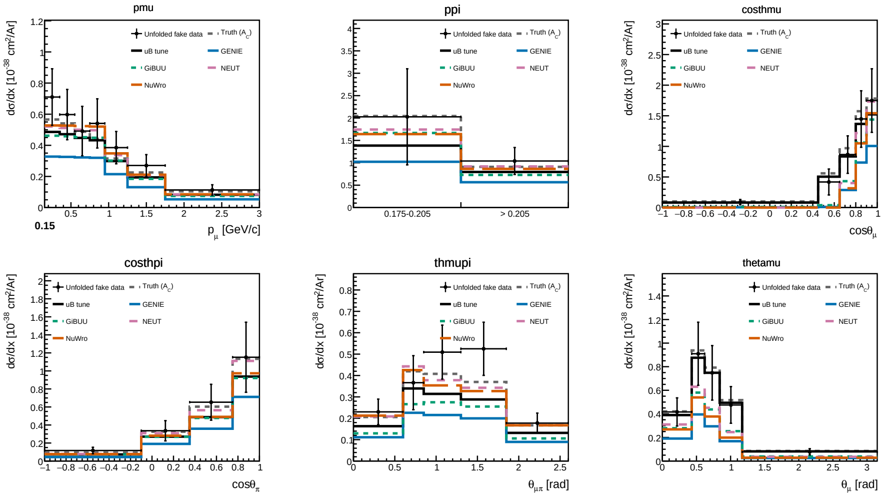
p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV/c).

Differential cross sections, FHC: + NuWro



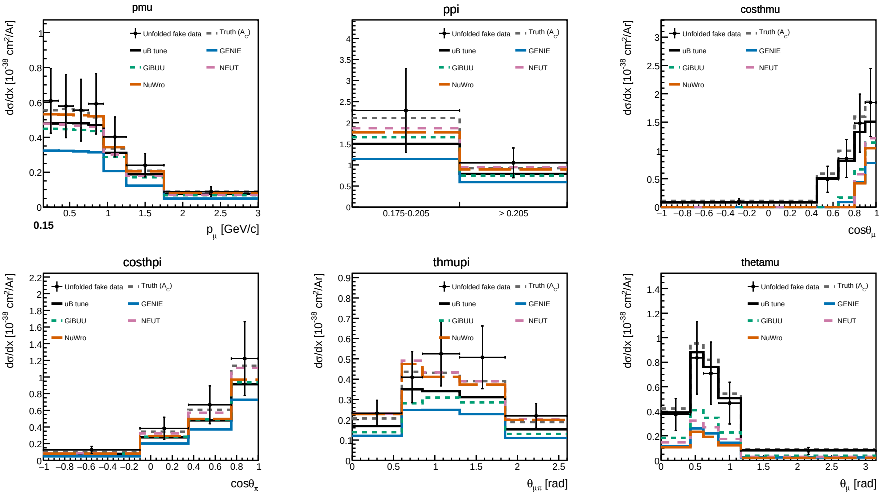
All four generators bracket the data. Error bars are stat ⊕ syst: the full covariance (flux, cross-section, detector, re-interaction, POT, targets, data/MC/EXT stats) propagated through the unfolding, $\simeq$27% for p_μ (FHC), flux-dominated (23%). p_π uses

Differential cross sections, RHC



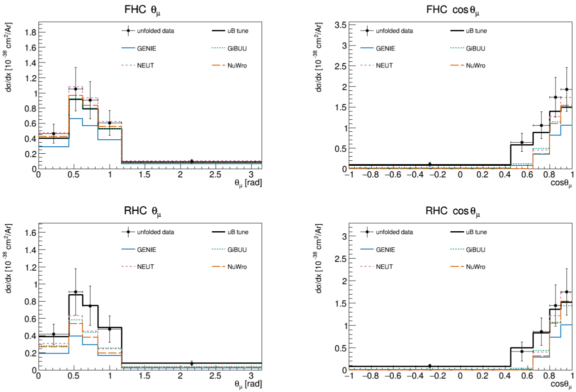
$\bar{\nu}_\mu$-enriched beam; unfolded fake data vs A_C-smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro. p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV/c).

Differential cross sections, combined



FHC+RHC combined exposure (per-universe correlated flux treatment). Error bars are stat $\oplus$ syst (full covariance). p_π uses the adopted two-bin scheme, drawn equal-width (bins are 0.030 and 0.795 GeV$/c$).

Muon angle: $\cos\theta_\mu$ vs θ_μ



All curves A_C-smeared (same space as the data). Same events, same binning (θ edges $=$ arccos of the $\cos\theta$ edges); only the variable differs.

$\cos\theta_\mu$ vs θ_μ: the FHC θ_μ “agreement” is an artefact

FHC θ_μ: normalised shapes

	bin 1	2	3	4	5
data	.1485	.3361	.2899	.1934	.0321
truth	.1484	.3359	.2901	.1933	.0323
uB tune	.1484	.3359	.2901	.1934	.0322
NEUT	.1484	.3359	.2901	.1934	.0322

A_C 2nd/1st singular value
(same 4 generator inputs, $s_2/s_1 = 5.0\times10^{-2}$):

FHC θ_μ	2.8×10^{-6}
FHC $\cos\theta_\mu$	5.1×10^{-2}
FHC p_μ	6.6×10^{-3}
RHC θ_μ	2.3×10^{-2}

- Retraction. We previously read the FHC θ_μ χ^2 drop ($\sim$26/5 $\to\sim$2.8/5) as evidence that θ_μ cures the $\cos\theta_\mu$ tension. It does not.
- In FHC θ_μ the data, the truth, the tune and all four generators share one shape to 4 decimals. A_C there is rank-1: it annihilates every shape degree of freedom, so each model is mapped onto the same curve and only its normalisation survives.
- $\Rightarrow$ the agreement is by construction, effective $\mathrm{ndf}\approx 1$, and $\chi^2/5$ is not a goodness-of-fit. The same four generator inputs are full rank ($s_2/s_1 = 5\times10^{-2}$), and $\cos\theta_\mu/p_\mu$/RHC θ_μ all retain their shape modes, so this is over-regularisation specific to FHC θ_μ, not a property of θ.
- The earlier “A_C penalises the forward peak” explanation is also wrong: across observables A_C vs peakedness gives $r=-0.17$ ($W_{\pi p}$ sharpest yet 0.78; p_n flat yet enhanced 1.71), and within $\cos\theta_\mu$ the flattest models (NEUT, NuWro) are suppressed most.
- Action: FHC θ_μ withdrawn from model comparison pending a regularisation re-run. All 18 extractions were re-unfolded with A_C now dumped to file, FHC θ_μ is the only rank-1 case (backup).

Backup: how much model discrimination survives A_C (all 18)

Beam	Obs	before	after	keep
FHC	p_μ	.0248	.0052	21%
FHC	p_π	.0586	.0080	14%
FHC	$\cos\theta_\mu$	.0554	.0724	131%
FHC	$\cos\theta_\pi$	.0589	.0211	36%
FHC	$\theta_{\mu\pi}$	.0422	.0391	93%
FHC	θ_μ	.0554	2.5e-6	**0%**
RHC	p_μ	.0232	.0036	15%
RHC	p_π	.0556	.0068	12%
RHC	$\cos\theta_\mu$	.0644	.0771	120%
RHC	$\cos\theta_\pi$	.0629	.0165	26%
RHC	$\theta_{\mu\pi}$	.0505	.0295	58%
RHC	θ_μ	.0644	.0188	29%
COMB	p_μ	.0240	.0028	12%
COMB	p_π	.0571	.0064	11%
COMB	$\cos\theta_\mu$	.0598	.5500	919%
COMB	$\cos\theta_\pi$	.0609	.0166	27%
COMB	$\theta_{\mu\pi}$	.0463	.0360	78%
COMB	θ_μ	.0598	.0070	12%

- "before/after" = largest pairwise difference between normalised generator shapes, truth-level vs A_C-smeared. Keep $\approx 0 \Rightarrow$ the measurement can no longer tell the models apart.
- FHC θ_μ is the only zero; the only A_C whose rows are all one vector (spread 9%, eff. rank 1.02; everything else 170–1050%, 1.24–1.91). RHC/COMB θ_μ are not affected.
- Momenta keep only 11–21% everywhere: not destroyed, but a good χ^2 on p_μ/p_π is a weak statement. $\theta_{\mu\pi}$ (58–93%) is the healthiest variable we have.
- The $>$ 100% for $\cos\theta_\mu$ is not good news: A_C pushes the generators negative in the backward bins while the tune stays positive (COMB bin 2: NuWro -0.43 vs tune $+0.51$). Unphysical $\Rightarrow$ that spread is artefact, not discrimination, and its χ^2 inherits the problem.
- So the original framing was backwards: neither muon-angle variable is clean, θ_μ loses everything (FHC), $\cos\theta_\mu$ keeps shape only via negative excursions.

Integrated & total flux-averaged cross section

Integrated σ_{int} (Wiener-SVD)

	FHC	RHC	Comb
p_μ	0.99	0.86	0.82
p_π	0.96	0.97	0.96
$\cos\theta_\mu$	0.82	0.66	0.70
$\cos\theta_\pi$	0.97	0.81	0.86
$\theta_{\mu\pi}$	0.88	0.88	0.92
θ_μ	1.00	0.83	0.78
θ_π	0.97	0.80	0.86

spread = Wiener-SVD A_C smearing ($10^{-38}\text{cm}^2/\text{Ar}$)

Total (cut-and-count, no unfolding)

Config	σ ($10^{-38}\text{cm}^2/\text{Ar}$)
FHC	1.06 ± 0.27
RHC	0.98 ± 0.35
Combined	1.02 ± 0.30

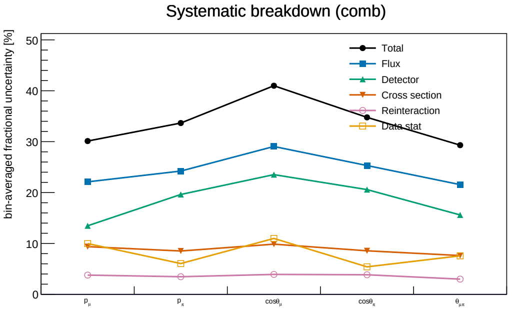
Uncertainties are syst $\oplus$ stat (e.g. FHC: 25.5% systematic, 4.3% data-statistical). Cut-and-count carries no A_C, so it is directly comparable to the truth-level generator column.

Generator totals: the spread is NOT A_C

	truth-level	A_C-smeared	A_C factor
GENIE	0.82	0.56	0.69
GiBUU	1.13	0.78	0.69
NEUT	1.32	0.84	0.64
NuWro	1.25	0.90	0.72

A_C is a common $\simeq$0.7 factor on all four models, so it cannot generate the $1.6\times$ spread between them at truth level; that is a

Systematics & sideband validation



Flux (PPFX) dominates; detector second; total $\sim$ 30–40%.

Background-control sidebands (combined)

Region	N_{data}	EXT	sig%	data/νMC
CC0π	24 148	17 855	9.7	1.00
π^0	14 694	10 284	9.9	1.01
multi-π	303	83	22.9	1.08
cosmic	169	616	8.0	1.10

Each inverts one signal requirement. CC0π $\rightarrow$ the FSI absorption path in/out of the signal; π^0 $\rightarrow$ shower-veto efficiency; multi-π $\rightarrow$ wrong-topology feed-down; cosmic $\rightarrow$ the only EXT-dominated region (4.0$\times$ the νMC).

Pseudo-data is a throw of the νMC, so data/νMC tests the machinery (per-run POT + weighting), not yet the background model. Small-region scatter is Poisson ($\sqrt{N}\approx$8% for cosmic).

At unblinding: CC0π/π^0 constrain to $\sim$1%, well inside the 20–30% xsec systematic they test.

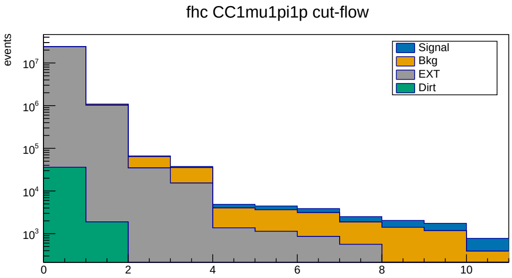
Part II, Proton-tagged subsample: hadronic mass & TKI

Idea

Requiring an identified proton reconstructs the full hadronic system, opening two new observable classes: the hadronic invariant mass (probes Δ/N^* directly) and the transverse kinematic imbalance (Fermi motion + FSI).

- `CC1mu1pi1p`: inherits the inclusive selection + a leading proton (LLR PID, $p_p \geq 0.3$ GeV$/c$, contained). Efficiency 11.5%, purity 48%.
- Six W/TKI observables: $W_{\pi p}$, W_{had}, δp_T, $\delta\alpha_T$, $\delta\phi_T$, p_n (6 bins each) + five inclusive kinematic observables re-measured on the proton-tagged phase space.

Selection performance & proton-tag quality



FHC proton-tagged cut-flow; last stage = proton identification

Proton tag (truth-matched, FHC)

- 87% tag purity ($\sim$ 13% mis-tag)
- Rejects $\sim$ 64% background for $\sim$ 30% signal loss
- Purity 33% $\rightarrow$ 48%; $S/\sqrt{B}$ +17%
- Proton p: -9% bias, 21% resolution

Proton-tagged cut-flow, FHC (full exposure)

Stages None–Nonprotons are the inherited inclusive selection (with the proton-tagged signal definition: inclusive signal + a true proton > 0.3 GeV/c); the final **IdentProton** stage adds the leading-proton identification that defines the subsample.

Cut	Signal	Bkg	EXT	Pred	Eff%	Pur%
None	3 387	281 372	23.92M	24.7M	100.0	0.01
InFiducialVol	2 834	63 944	1.02M	1.11M	83.7	0.26
Topological	2 238	29 303	34 481	66 929	66.1	3.3
MuonCandidate	2 031	20 092	15 206	37 776	60.0	5.4
ContainedPion	852	2 637	1 348	4 890	25.2	17.4
MuonIn3Planes	812	2 499	1 122	4 473	24.0	18.2
PionIn3Planes	754	2 255	854	3 888	22.3	19.4
ShowerCut	632	1 313	558	2 521	18.7	25.1
OpeningAngle	628	1 226	186	2 045	18.5	30.7
Nonprotons	568	1 049	128	1 749	16.8	32.5
IdentProton	**401**	**373**	**46**	**828**	**11.8**	**48.4**

IdentProton rejects $\sim$64% of the ν background while keeping $\sim$71% of signal: purity 32.5% $\rightarrow$ 48.4%. RHC/comb: 46.6%/47.5% purity at $\sim$12% efficiency.

The six proton-tagged observables

Built from the muon, charged-pion and leading-proton four-vectors ($\vec{p}^{\,\mathrm{had}} = \vec{p}_\pi + \vec{p}_p$, transverse components w.r.t. the beam $\hat{z}$; framework STV tool):

Hadronic mass, probes Δ/N^*

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}$$

$$W_{\mathrm{had}} = \sqrt{m_p^2 + 2m_p(E_{\mathrm{cal}} - E_\mu) - Q^2}$$

$W_{\pi p}$ is directly measured; no assumption on the ν direction or energy. W_{had} is calorimetric, $E_{\mathrm{cal}} = E_\mu + E_\pi + T_p$.

Threshold: $W_{\pi p} \geq m_\pi + m_p = 1.08$ GeV$/c^2$.

TKI, probes Fermi motion & FSI

$$\delta p_T = |\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\mathrm{had}}|$$

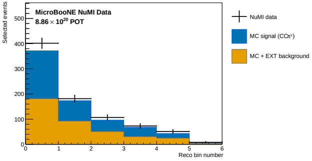
$$\delta\alpha_T = \angle(-\vec{p}_T^{\,\mu},\ \vec{p}_T^{\,\mu} + \vec{p}_T^{\,\mathrm{had}})$$

$$\delta\phi_T = \angle(-\vec{p}_T^{\,\mu},\ \vec{p}_T^{\,\mathrm{had}})$$

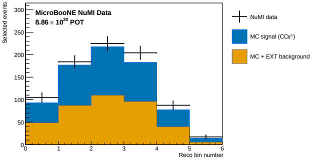
$$p_n = \sqrt{\delta p_T^2 + p_L^2}$$

With no nuclear effects $\delta p_T \to 0$; Fermi motion broadens it and FSI drive $\delta\alpha_T$ toward 180°. p_n infers the initial-state nucleon momentum ($E_b = 24.78$ MeV argon removal energy enters p_L).

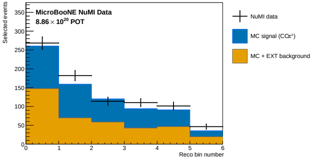
Each is measured in six equal-population bins; the proton is required contained with $|\vec{p}_p| > 0.3$ GeV$/c$.



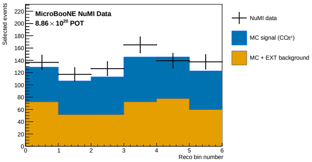
$W_{\pi p}$



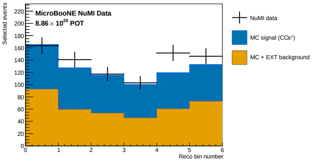
W_{had}



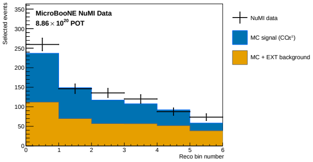
δp_T



$\delta\alpha_T$



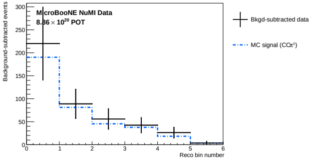
$\delta\phi_T$



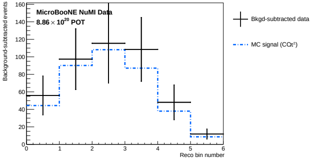
p_n

Points: (blind) fake data; stack: MC signal + background. Purity at the IdentProton stage is 48%, lower than the inclusive selection, so the subtraction is proportionally larger.

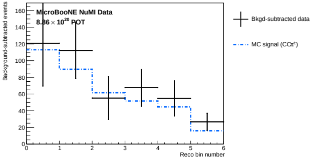
Proton-tagged, FHC: after background subtraction



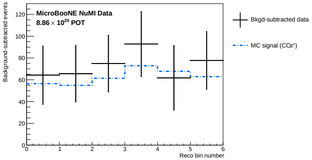
$W_{\pi p}$



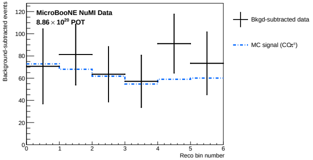
W_{had}



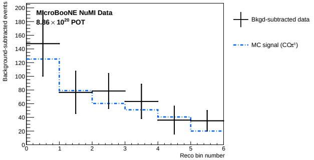
δp_T



$\delta\alpha_T$



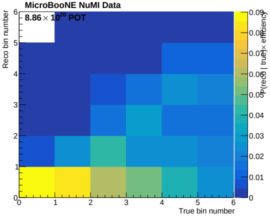
$\delta\phi_T$



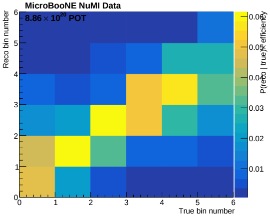
p_n

Data − estimated background vs CV MC signal; the input to the unfolding for the six W/TKI observables.

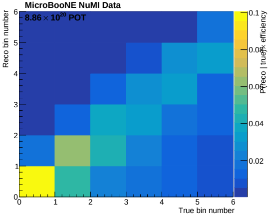
Proton-tagged: response (smearceptance) matrices, FHC



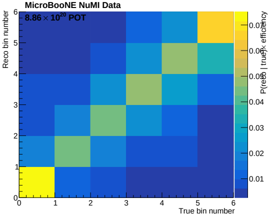
$W_{\pi p}$



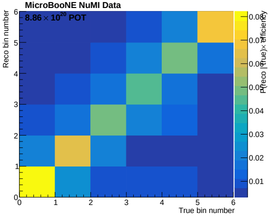
W_{had}



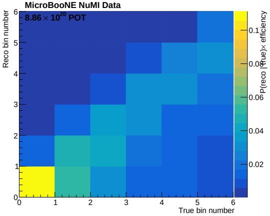
δp_T



$\delta\alpha_T$



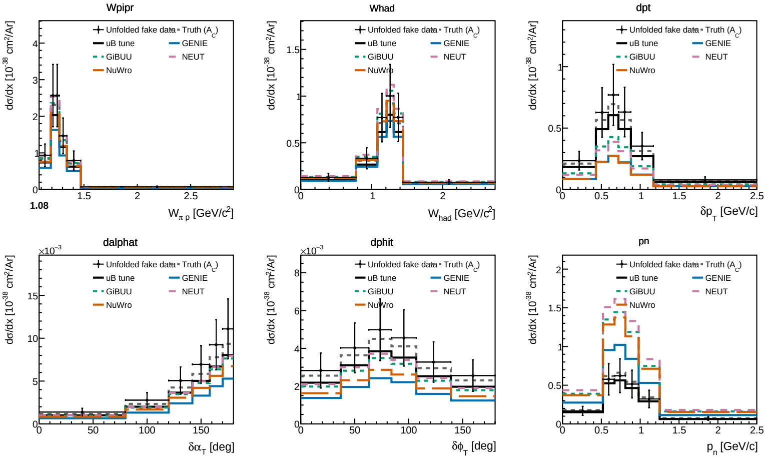
$\delta\phi_T$



p_n

A_C: W_{had} 0.88, $W_{\pi p}$ 0.78, $\delta\alpha_T$ 0.63, $\delta\phi_T$ 0.61, δp_T 0.43, p_n 1.71; χ^2 tracks $|A_C - 1|$ (backup).

Proton-tagged differential cross sections (FHC)



Unfolded fake data vs A_C-smeared truth, the uB tune and all four generators (predictions regenerated at the 6-bin analysis binning). Closure $\chi^2/\mathrm{ndf} \ll 1$ for every observable.

Proton-tagged closure: all three beam configurations

Observable	$\sigma_{\rm int}$ [10^{-38}cm^2/Ar] FHC	RHC	COMB	COMB χ^2/ndf
$W_{\pi p}$	0.62	0.578	0.564	0.07/6
$W_{\rm had}$	0.60	0.500	0.504	1.30/6
δp_T	0.61	0.493	0.497	0.71/6
$\delta\alpha_T$	0.67	0.567	0.667	0.86/6
$\delta\phi_T$	0.63	0.568	0.573	0.05/6
p_n	0.52	0.548	0.556	1.00/6
p_μ	0.56	0.531	0.523	0.65/7
p_π	0.60	0.572	0.582	2.49/5
$\cos\theta_\mu$	0.52	0.512	0.529	0.52/5
$\cos\theta_\pi$	0.54	0.444	0.454	2.17/4
$\theta_{\mu\pi}$	0.50	0.410	0.447	0.21/5

- All eleven observables now close positive in all three configurations, the proton-tagged measurement is complete. Every combined χ^2 is well below its number of bins.
- The RHC and combined results were previously negative and withdrawn. The cause was a POT-normalisation bug, not the data: the processed ntuples carry two POT conventions, and normalising by the sum of the distinct per-file exposures is correct for both.
- COMB lies between FHC and RHC for 7 of 11. The four exceptions are small (p_n $+6.9\%$ vs FHC; $\cos\theta_\mu$ $+1.7\%$; $W_{\pi p}$ -2.4% and p_μ -1.5% vs RHC), COMB is an independent extraction with its own response, regularisation and A_C, not a fluence-weighted average, so it is not required to bracket. All four are far inside the $\sim 30\%$ systematic.
- The 0.45–0.67 spread across observables is A_C, not physics: each observable's smearing rescales the integral differently, so these are not eleven independent measurements of one number.

Generator comparison & background composition

Integrated proton-tagged σ ($10^{-38}\text{cm}^2/\text{Ar}$)

Model	σ (truth level)
GENIE	0.50
NuWro	0.65
GiBUU	0.74
NEUT	0.78
Unfolded data ($W_{\pi p}$)	0.62

Spread $\sim 1.6\times$, largest in low-δp_T / high-p_n (FSI-sensitive). Generator values are truth-level (no A_C); predictions regenerated at the 6-bin analysis binning, NuWro and NEUT from event vectors in the SL7 container.

Remaining background (FHC)

Component	%
ν_μ CC wrong topology	64
True CC$1\mu1\pi$, no $p>0.3$	18
NC	11
Out-of-FV	5
ν_e CC	1

82% true ν_μ CC , well modelled & constrainable.

Proton-tagged: background-control sidebands

The proton-tagged selection inherits the four sideband definitions from `CC1mu1piXp`; no new regions or reprocessing needed. Each can be read either as inherited, or with a tagged proton additionally required (the measurement phase space).

Region (comb)	as inherited N_{data}	EXT	sig%	d/νMC	+ proton tag N_{data}	EXT	sig%	d/νMC
CC0π	24 148	17 855	9.7	1.00	16 555	2 891	6.0	1.01
π^0	14 694	10 284	9.9	1.01	9 019	2 370	7.1	1.02
multi-π	303	83	22.9	1.08	137	36	18.9	1.07
cosmic	169	616	8.0	1.10	60	12	7.1	1.07

- More signal-depleted than for the inclusive selection (6–7% vs 48% purity in the signal region).
- The proton tag costs $\sim 1/3$ of the statistics but cuts EXT by $\sim 6\times$, cosmics rarely have a reconstructed proton.
- $\Rightarrow$ the EXT normalisation systematic matters much less here than in the inclusive measurement.
- Gap: these *apply* the proton tag, so none constrains the $\sim$13% proton mis-tag. That needs an inverted-PID region (new branch + reprocess).

Part III, Multi-pion: exclusive CC $1\mu2\pi$ and CC $1\mu3\pi$

New exclusive channels, building on the work of **Tusharadri Mahmud** (MSc) and **Linhui Gu**. Thin subselections of the inclusive selection: same muon, but *exactly* 2 or 3 charged pions (any number of protons). They probe the resonant/DIS transition and the intranuclear pion cascade, and are a background to the single-pion signal.

Tuning vs. the 1π selection

Setting	1π	Multi-π
Pion–vertex distance	$\leq$ 4 cm	$\leq$ 9.5 cm (looser)
Uncontained pions	0	up to 2
Opening-angle cut	on	off
Shower veto	on	off
Wire-gap cuts	on	off
Per-pion threshold	0.175	0.10 GeV/c
Pion ID	LLR	dedicated BDT

Performance (Run 1 FHC only)

	2π	3π
Efficiency	17.9%	10.1%
Purity	26.8%	11.3%
Events @ full exposure	$\sim$670	$\sim$66
Reco ceiling	66.6%	52.4%

Efficiency/purity measured on the Run-1 FHC ν_μ overlay alone (one MC file); event yields are extrapolated to the full FHC exposure. RHC is not yet processed for the multi-π channels.

The reco ceiling (fraction of true pions reconstructed as tracks at all) shows the loss is particle-ID, not reconstruction.

Multi-pion: dedicated pion-ID BDT

- Inherited loose LLR cut replaced by a dedicated BDT (TMVA gradient boost), trained to separate true charged pions from the tracks polluting the multi-π sample. ROC $= 0.81$.
- Inputs: pion/proton LLR, Bragg p/μ/MIP likelihoods, track score, length, vertex distance; no explicit momentum.
- 2π uses two momentum-specialised BDTs (split at 20 cm track length, tighter soft-pion cut to protect the exact-multiplicity purity); 3π keeps a single BDT (its pions are genuinely soft).
- Trialled on the inclusive selection it hurts (eff $16.7 \rightarrow 18.9\%$ but purity $60.3 \rightarrow 38.8\%$) $\Rightarrow$ inclusive keeps its LLR cut.

Residual background (truth-matched)

Counted "pion" really is	%
real out-of-topology $\pi^{\pm}$	44
proton mis-ID	31
muon	12
other	13

Proton contamination is the irreducible core; a proton ranging out like a pion is genuinely ambiguous; adding a Bragg veto on top of the BDT did *not* reduce it and only cost efficiency.

Documented in a standalone technote; differential 2π measurement + total σ for 3π are the targets.

Summary

- Inclusive: differential CC1μ1$\pi^{\pm}$ cross sections for FHC, RHC and combined, per-run POT scaling, Wiener-SVD unfolding; total flux-averaged σ agrees with GENIE/GiBUU/NEUT/NuWro within $\sim 1\sigma$.
- Proton-tagged: new hadronic-mass ($W_{\pi p}$, W_{had}) and TKI (δp_T, $\delta\alpha_T$, $\delta\phi_T$, p_n) observables, plus the five inclusive kinematic observables re-measured. Closure validated in all three configurations at six-bin granularity, eleven observables $\times$ FHC, RHC and combined, all positive, every combined χ^2 below its bin count.
- Blind analysis: fake data throughout; sidebands (CC0π, multi-π, π^0, cosmic) validated in all three beams, data/νMC $\simeq 1.00$, ready to constrain the background before unblinding.
- p_π scope change: its response fails the standard binning criterion ($>$68% diagonal) in 4 of 5 bins, and a trained golden-pion BDT, even at perfect truth-level selection, does not repair it. We expect to quote p_π on the golden-pion-enhanced and/or proton-tagged samples rather than the inclusive one. No other observable is affected.
- Multi-pion: an analysis of the exclusive 2π and 3π channels has started, building on the work of Tusharadri Mahmud (MSc) and Linhui Gu.

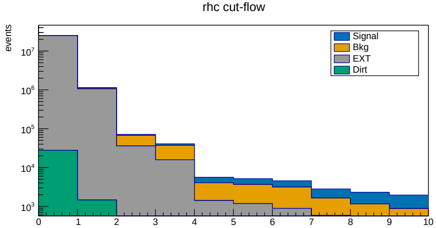
Thank you, questions?

Backup

Backup: inclusive cut-flow, RHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	6 131	304 347	24.84M	100.0	0.02
InFiducialVol	5 124	69 102	1.06M	83.6	0.44
Topological	3 960	31 629	35 801	64.6	5.5
MuonCandidate	3 559	21 443	15 788	58.0	8.6
ContainedPion	1 577	2 649	1 400	25.7	27.8
MuonIn3Planes	1 512	2 500	1 165	24.7	29.0
PionIn3Planes	1 409	2 258	887	23.0	30.8
ShowerCut	1 171	1 061	579	19.1	41.4
OpeningAngle	1 162	965	193	19.0	50.0
Nonprotons	**1 081**	**746**	**133**	**17.6**	**55.0**

RHC (Runs 1,2,3,4a-c; 1.108×10^{21} POT). MC prediction, per-run POT scaled; data blind.

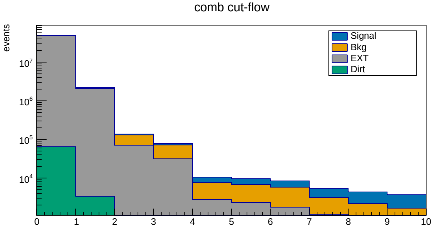


RHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale).

Backup: inclusive cut-flow, combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	11 710	583 527	48.76M	100.0	0.02
InFiducialVol	9 756	131 249	2.08M	83.3	0.43
Topological	7 461	59 669	70 282	63.7	5.4
MuonCandidate	6 702	40 423	30 994	57.2	8.5
ContainedPion	2 972	4 743	2 749	25.4	28.2
MuonIn3Planes	2 847	4 475	2 287	24.3	29.4
PionIn3Planes	2 648	4 028	1 742	22.6	31.3
ShowerCut	2 218	1 959	1 137	18.9	41.5
OpeningAngle	2 202	1 778	379	18.8	50.4
Nonprotons	**2 055**	**1 389**	**261**	**17.5**	**55.4**

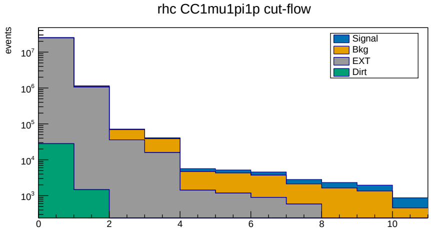
Combined (1.994×10^{21} POT). Efficiency/purity track the two beams closely; the selection is beam-independent.



Combined stack (log scale).

Backup: proton-tagged cut-flow, RHC (full exposure)

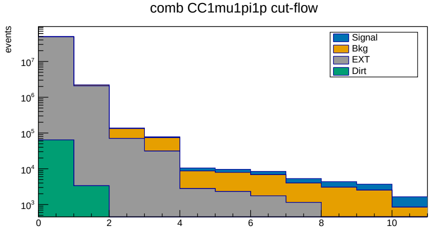
Cut	Signal	Bkg	EXT	Eff%	Pur%
None	3 638	306 840	24.84M	100.0	0.01
InFiducialVol	3 064	71 162	1.06M	84.2	0.26
Topological	2 455	33 134	35 801	67.5	3.4
MuonCandidate	2 234	22 769	15 788	61.4	5.4
ContainedPion	935	3 291	1 400	25.7	16.5
MuonIn3Planes	892	3 119	1 165	24.5	17.1
PionIn3Planes	831	2 835	887	22.9	18.2
ShowerCut	687	1 544	579	18.9	24.3
OpeningAngle	683	1 444	193	18.8	29.4
Nonprotons	611	1 217	133	16.8	31.1
IdentProton	**436**	**443**	**48**	**12.0**	**46.6**



RHC proton-tagged stack; last stage IdentProton (leading-proton tag). Purity 31.1% → 46.6%.

Backup: proton-tagged cut-flow, combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	7 025	588 212	48.76M	100.0	0.01
InFiducialVol	5 898	135 106	2.08M	84.0	0.26
Topological	4 693	62 436	70 282	66.8	3.4
MuonCandidate	4 265	42 860	30 994	60.7	5.4
ContainedPion	1 787	5 928	2 749	25.4	16.9
MuonIn3Planes	1 704	5 618	2 287	24.3	17.6
PionIn3Planes	1 585	5 091	1 742	22.6	18.7
ShowerCut	1 320	2 857	1 137	18.8	24.7
OpeningAngle	1 310	2 670	379	18.7	30.0
Nonprotons	1 179	2 266	261	16.8	31.8
IdentProton	**837**	**816**	**95**	**11.9**	**47.5**



Combined proton-tagged stack; IdentProton purity $31.8\% \to 47.5\%$.

Backup: per-run exposure

Mode	Run	Data POT	MC/data scale
FHC	1	3.28×10^{20}	0.141
	2	1.27×10^{20}	0.051
	4	2.08×10^{20}	0.073
	5	2.23×10^{20}	0.116
RHC	1–4	1.11×10^{21}	0.045–0.091

Each run scaled by its own $D_{\mathrm{run}}/\mathrm{MC}_{\mathrm{run}}$ (real-data-ready).

Backup: systematic breakdown (combined)

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	30.1	33.7	41.0	34.8	29.3
Flux (PPFX)	22.1	24.2	29.1	25.3	21.6
Detector	13.5	19.6	23.5	20.6	15.6
Cross section	9.4	8.5	9.9	8.6	7.6
Data stat	10.0	6.1	11.0	5.4	7.6

Bin-averaged fractional uncertainty [%]. Flux dominates throughout.

Backup: history; the superseded pion-momentum estimator

An earlier version of this analysis reconstructed the pion momentum from the proton-hypothesis range kinetic energy (`trk_energy_proton_v`) rather than the muon-hypothesis range momentum used now.

- That is a kinetic energy, while the true axis it was binned against (with identical bin edges) is a momentum, two different physical quantities.
- Consequence: the migration matrix was systematically shifted rather than near-diagonal. A true 0.2 GeV/c pion gives $KE \approx 0.104$ GeV, two bins low; a true 0.175 GeV/c pion falls below the first reco bin entirely.
- Fixed by switching to `trk_range_muon_mom_v` (`CC1mu1piXp.cxx`), which returns a momentum and exploits $m_\mu \approx m_\pi$.

Note: the proton-hypothesis range KE remains the *correct* estimator for the proton in the W/TKI subsample; there the quantity wanted really is a kinetic energy, converted to momentum via $|\vec{p}_p| = \sqrt{T_p^2 + 2T_p m_p}$.

Resolution panels produced before the switch quote a $\sim$21% underestimate of p_π; they are to be regenerated with the current estimator.

Backup: from PeLEE branches to the kinematic inputs

Every observable is built from three reconstructed tracks (muon, charged pion, leading proton), each taken from the PeLEE `NeutrinoSelectionFilter` ntuple at the candidate's PFP index:

Quantity	PeLEE branch	Note
$\lvert\vec{p}_\mu\rvert$	`trk_range_muon_mom_v`	range-based (primary)
$\lvert\vec{p}_\mu\rvert$ (x-check)	`trk_mcs_muon_mom_v`	multiple-scattering
$\lvert\vec{p}_\pi\rvert$	`trk_range_muon_mom_v`	μ-hypothesis range, at the π index
$T_p \to \lvert\vec{p}_p\rvert$	`trk_energy_proton_v`	p-hypothesis range KE; $\lvert\vec{p}_p\rvert = \sqrt{T_p^2 + 2T_p m_p}$
$\hat{u}_\mu, \hat{u}_\pi, \hat{u}_p$	`trk_dir_x/y/z_v`	unit track direction
$\cos\theta$	from $\hat{u}_z$	w.r.t. detector z (beam-ish) axis
$\theta_{\mu\pi}$	$\arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$	opening angle
PID / candidacy	`trk_llr_pid_score_v`, `trk_bragg_*_v`, `trk_score_v`, `trk_distance_v`, `trk_len_v`	see PID slide

Energies follow as $E_i = \sqrt{\lvert\vec{p}_i\rvert^2 + m_i^2}$ with the PDG mass of the assigned species. Truth counterparts use `mc_nu_daughter_p[xyz]` (back-matched via `backtracked_pdg`).

Backup: W/TKI observable definitions

From the muon, charged-pion and leading-proton four-vectors above (framework STV tool):

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2} \quad \text{(direct, no } \nu \text{ direction)}$$

$$W_{\mathrm{had}} = \sqrt{m_p^2 + 2m_p(E_{\mathrm{cal}} - E_\mu) - Q^2} \quad \text{(calorimetric)}$$

$$\delta p_T = |\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\mathrm{had}}|, \quad \delta\alpha_T,\ \delta\phi_T \text{ (TKI angles)}, \quad p_n = \sqrt{\delta p_T^2 + p_L^2}$$

with $\vec{p}^{\,\mathrm{had}} = \vec{p}_\pi + \vec{p}_p$, $E_{\mathrm{cal}} = E_\mu + E_\pi + T_p$, and the transverse components taken w.r.t. the beam ($\hat{z}$) axis. $E_b = 24.78$ MeV argon removal energy enters p_L (hence p_n).

Backup: proton-tagged binning & closure

Observable	Bin edges (6 equal-population bins)
$W_{\pi p}$	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.00 GeV/c^2
W_{had}	0, 0.78, 1.08, 1.21, 1.31, 1.44, 1.90 GeV/c^2
δp_T	0, 0.43, 0.59, 0.72, 0.88, 1.17, 1.60 GeV/c
$\delta\alpha_T$	0, 80, 120, 142, 158, 170, 180 deg
$\delta\phi_T$	0, 37, 63, 84, 107, 139, 180 deg
p_n	0, 0.52, 0.67, 0.81, 0.98, 1.25, 1.70 GeV/c

$+$ five kinematic observables (p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) on inclusive binning.

FHC W/TKI closure σ_{int}: 0.62, 0.60, 0.61, 0.67, 0.63, 0.52; $\chi^2/\mathrm{ndf} \ll 1$, all positive.
RHC closure also validated: all 11 observables positive, σ_{int} 0.41–0.58, just below the FHC band as expected for the $\bar{\nu}_\mu$-enriched flux. Combined in production.

A_C **smearing factor** (smeared/truth): W_{had} 0.88, $W_{\pi p}$ 0.78 (sharpest distribution, yet barely affected, no W reparameterisation needed), $\delta\alpha_T$ 0.63, $\delta\phi_T$ 0.61, δp_T 0.43, p_n 1.71 (enhanced). The last two distort model comparison in those panels.

Backup: χ^2/ndf, inclusive, all observables and beams

Unfolded fake data compared with each A_C-smeared prediction.

Beam	Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	1.66/7	3.09/7	5.99/7	3.87/7	3.77/7	2.90/7
FHC	p_π	1.96/5	3.77/5	5.05/5	3.78/5	3.33/5	3.46/5
FHC	$\cos\theta_\mu$	0.47/5	3.62/5	9.67/5	12.43/5	26.28/5	26.02/5
FHC	$\cos\theta_\pi$	0.49/4	1.20/4	3.82/4	0.99/4	1.34/4	1.19/4
FHC	$\theta_{\mu\pi}$	0.59/5	2.59/5	4.12/5	3.24/5	2.47/5	2.29/5
RHC	p_μ	5.27/7	8.57/7	10.88/7	8.81/7	8.15/7	8.48/7
RHC	p_π	6.09/5	7.04/5	9.36/5	7.28/5	5.96/5	6.13/5
RHC	$\cos\theta_\mu$	0.96/5	4.91/5	7.45/5	8.66/5	18.43/5	17.42/5
RHC	$\cos\theta_\pi$	1.47/4	4.00/4	6.64/4	3.92/4	4.21/4	4.30/4
RHC	$\theta_{\mu\pi}$	4.32/5	5.57/5	7.67/5	5.68/5	6.56/5	7.59/5
Comb	p_μ	3.41/7	3.70/7	5.82/7	3.99/7	3.83/7	3.30/7
Comb	p_π	1.32/5	2.32/5	3.85/5	2.65/5	1.82/5	2.05/5
Comb	$\cos\theta_\mu$	2.49/5	4.61/5	9.07/5	12.02/5	26.72/5	27.20/5
Comb	$\cos\theta_\pi$	0.42/4	1.83/4	4.01/4	1.69/4	1.89/4	1.95/4
Comb	$\theta_{\mu\pi}$	2.10/5	2.42/5	3.91/5	3.05/5	3.49/5	3.45/5

The small *truth* column is the closure. RHC closure is visibly worse than FHC. The $\cos\theta_\mu$ row is an outlier for NEUT/NuWro, see next slide.

Backup: the $\cos\theta_\mu$ χ^2 is largely an A_C artefact

Same events, same data, only the angular variable changes:

Beam	Variable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos\theta_\mu$	0.47/5	3.62/5	9.67/5	12.43/5	26.28/5	26.02/5
FHC	θ_μ	0.05/5	3.07/5	4.73/5	3.03/5	2.82/5	2.90/5
RHC	$\cos\theta_\mu$	0.96/5	4.91/5	7.45/5	8.66/5	18.43/5	17.42/5
RHC	θ_μ	0.48/5	4.41/5	6.97/5	5.54/5	8.18/5	8.46/5
FHC	$\cos\theta_\pi$	0.49/4	1.20/4	3.82/4	0.99/4	1.34/4	1.19/4
FHC	θ_π	0.44/4	1.15/4	3.77/4	0.93/4	1.30/4	1.18/4

In $\cos\theta_\mu$ the A_C smearing suppresses the peaked generators by $\times 0.33$ but the smoother tune by much less, so NEUT and NuWro are pushed far from the data and χ^2 explodes. In θ_μ, where the suppression is $\times 0.76$, the same models agree well (χ^2 falls by $\sim 9\times$). The π angle, never sharply peaked, is unchanged. The large $\cos\theta_\mu$ χ^2 is therefore mostly a variable-choice artefact, not model disagreement; a further argument for quoting θ rather than $\cos\theta$.

Backup: χ^2/ndf, proton-tagged (FHC)

Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro	A_C
W_{had}	0.09/6	1.39/6	1.65/6	1.06/6	1.17/6	1.05/6	0.88
$W_{\pi p}$	0.09/6	0.95/6	1.78/6	0.62/6	0.56/6	0.87/6	0.78
$\delta\alpha_T$	0.26/6	2.74/6	4.73/6	2.95/6	2.78/6	3.53/6	0.63
$\delta\phi_T$	0.09/6	5.14/6	7.21/6	5.61/6	5.39/6	6.47/6	0.61
δp_T	0.46/6	5.34/6	9.89/6	8.27/6	9.02/6	10.38/6	0.43
p_n	0.09/6	4.24/6	7.72/6	18.74/6	25.40/6	16.45/6	1.71

Rows are ordered by how close the additional-smearing factor A_C is to unity, and the χ^2 follows it monotonically: $|A_C - 1| = 0.12, 0.22, 0.37, 0.39, 0.57, 0.71$ against generator $\chi^2 \simeq 1.2, 1.0, 3.3, 6.2, 9.4, 17$. The two observables A_C barely touches (W_{had}, $W_{\pi p}$) agree with all four generators; the two it distorts most (δp_T suppressed to 0.43, p_n *enhanced* to 1.71) show the largest χ^2; in the same direction for every model. This is the signature of a smearing artefact, not model discrimination, the same effect seen in $\cos\theta_\mu$. The closure (truth column) is excellent throughout.

Backup: generator predictions

- Inclusive: per-observable $d\sigma/dx$ from GENIE (gst), GiBUU (`FinalEvents`), NEUT (neutvect), NuWro (event output), flux-combined to per-argon 10^{-38} cm^2.
- Proton-tagged W/TKI: same event samples reprocessed with the proton tag + TKI construction; NuWro and NEUT run inside the SL7 apptainer container.
- FTE rebinned to the coarsened analysis binning so every generator overlays on every observable.